

AI-Driven Delinquency Risk Report

Prepared for: Head of Collections, Geldium

Date: June 2026 Dataset: 500 customers × 19 variables Model: XGBoost (GBDT)

1. Summary of Predictive Insights

Analysis of Geldium's customer dataset — combined with a Gradient Boosted Decision Tree predictive model — reveals clear, actionable patterns in delinquency risk. Three variables dominate: how many payments a customer has missed, how much of their available credit they are using, and how their debt obligations compare to their income.

TOP 3 RISK FACTORS

- **1. Missed Payments** — Customers with 5–6 missed payments are overwhelmingly more likely to be delinquent. This is the single strongest behavioural signal in the dataset.
- **2. Credit Utilization >70%** — High utilization signals financial stress. Four records even exceed 100%, indicating balance overruns that correlate with payment failure.
- **3. Debt-to-Income Ratio >40%** — When debt obligations consume more than 40% of a customer's income, discretionary repayment capacity is severely constrained.

High-Risk Customer Segments

- **Unemployed customers** (93 records, 19% of dataset): Loss of income directly undermines repayment capacity, making this the highest-risk employment segment.
- **Customers with 4+ missed payments and Credit_Utilization >70%**: The combination of behavioural failure and financial stress creates a compounding risk profile.
- **High Debt-to-Income customers (>40%) with deteriorating payment history**: Sequential Late/Missed payments across Month_1–6 indicate a trajectory toward default, not a one-time event.

Noteworthy finding: Credit Score and Income level do not cleanly separate delinquent from non-delinquent customers in this dataset — delinquent customers actually show a marginally higher mean credit score (591 vs 575). Intervention strategies should therefore prioritise behavioural and capacity-based signals over bureau scores alone.

2. Recommendation Framework

Based on predictive findings, the highest-impact intervention targets customers showing **Credit_Utilization above 70%** — an early-warning signal that precedes missed payments and is actionable before accounts become delinquent.

SMART Criteria	Detail
Specific	Contact all customers with Credit_Utilization >70% and at least one missed payment in the last 6 months to offer a structured repayment plan.
Measurable	Target a 15% reduction in the delinquency rate within this segment over the next 90 days, tracked via monthly collections.

Achievable	Utilization and payment data are already captured in the dataset; no new data collection is required. Collections team
Relevant	Reducing delinquency in high-utilization accounts directly protects Geldium's loan book and reduces write-off exposure
Time-bound	Pilot the intervention for 90 days, review results at Day 30 and Day 60, with a full evaluation and decision on scale-up

For stakeholders: Our AI model identifies customers who are using most of their available credit as significantly more likely to miss future payments. By proactively reaching out to these customers with repayment support — before they default — Geldium can reduce write-offs and preserve customer relationships at the lowest possible intervention cost.

3. Ethical & Responsible AI Considerations

■ Fairness Risk 1: Geographic Proxy Bias

The *Location* field may act as a proxy for protected characteristics such as race or socioeconomic status, causing the model to penalise customers from certain postcodes unfairly.

Mitigation: Audit Location's SHAP contribution. If it acts as a demographic proxy, remove it from the feature set and replace with financially relevant geographic indicators (e.g., regional unemployment rate).

■ Fairness Risk 2: Employment Status Disparate Impact

Unemployed customers face higher predicted risk scores. If unemployment correlates with protected characteristics, the model may systematically disadvantage certain demographic groups — triggering fair lending concerns.

Mitigation: Calculate the Disparate Impact Ratio (DIR) across customer segments. A DIR below 0.80 mandates model review. Ensure *Employment_Status* drives predictions only via legitimate financial logic.

Explaining the Model to Non-Technical Stakeholders

The AI model works like an experienced collections analyst reviewing every customer's file simultaneously. It looks at the same information your team already uses — payment history, outstanding balances, income — and produces a **risk score from 0 to 1** for each customer. A score above the agreed threshold (e.g., 0.5) means the customer is flagged for proactive outreach. For every flagged customer, the model can show **the top three reasons** for their score (e.g., '4 missed payments, 85% credit utilization, high debt ratio'), so collections agents always understand *why* a customer was flagged — not just *that* they were.

Responsible AI Commitments

- ✓ **Transparency:** Every prediction is accompanied by a plain-language explanation of contributing factors via SHAP values.
- ✓ **Human oversight:** The model flags risk; trained collections agents make the final outreach decision.
- ✓ **Regular auditing:** Fairness metrics reviewed monthly; model retrained on rolling data to prevent drift.